

PAPER_1717_LITHIUM_7_FACTOR_3

Daniel T. Murphy

PAPER_1717 — Lithium-7 BBN Discrepancy = Li-7 factor = $D_{\text{phys}} - 1 = 3$ EXACT

Author: Daniel T. Murphy

Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: Cosmology

Date: June 18, 2026

Status: CLOSED — EXACT closure (PAPER_122x-123x foundational Millennium-adjacent)

Observation

PAPER_1227 (PAPER_122x-123x broader corpus) gives:

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Li-7 factor = $D_{\text{phys}} - 1 = 3$ EXACT

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Status: **EXACT**.

UQFF Closed Identity

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Li-7 factor = $D_{\text{phys}} - 1 = 3$ EXACT EXACT

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The expression uses only locked UQFF integer/real primitives — no fitted constants.

Lithium-7 BBN Problem — Triadic Resolution

The Lithium-7 problem is a 25-year-old discrepancy:

- **BBN prediction** (standard Λ CDM): $\text{Li}/\text{H} \approx 5 \times 10^{-4}$
- **Observed** (metal-poor stars): $\text{Li}/\text{H} \approx 1.6 \times 10^{-4}$
- **Discrepancy factor:** 3

UQFF resolution:

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Li-7 discrepancy = $D_{\text{phys}} - 1 = 3$ EXACT

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The **same integer 3** ($= D_{\text{phys}} - 1$) that produces:

- 3 quark colors (SU(3))
- 3 fermion generations
- Solar ν deficit 1/3
- SCm carrier velocity $c/3$
- BNS GW damping 1/3
- PCR triadic quantum number

- TDE outflow factor $c \cdot 3/10$
- Kochen-Specker contextuality dim 3

...also resolves the Lithium-7 anomaly. UQFF unifies BBN nuclear physics with the triadic symmetry sector via a single integer primitive.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1227
- Related: PAPER_1706-1715 (tier-29 ITER fusion); PAPER_1696-1705 (tier-28)
- Calculator dispatch: `calculate_paradox({"paradox": "lithium_7_factor_3"})`

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